

NGO Resource Allocation Playbook

Assam Flood Relief Forecasting System

Purpose

This system helps NGOs and disaster-response agencies anticipate flood-relief demand before major flood events occur. By combining rainfall forecasts, demographic indicators, historical flood vulnerability, and accessibility constraints, the platform estimates district-level ration-kit requirements for the upcoming 7 days.

Operational Workflow

Step 1: Review Daily Forecast Dashboard

Every morning, operations teams should review:

- Forecast rainfall (next 7 days)
- Predicted ration-kit demand
- District risk classification
- Recommended action level

Priority should be given to districts showing High or Severe risk.

Risk-Level Interpretation

Low Risk (< 2,000 kits)

Characteristics:

- Limited rainfall forecast
- Low expected disruption

Actions:

- Maintain existing stock
 - Continue monitoring
 - No immediate deployment required
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Medium Risk (2,000–5,000 kits)

Characteristics:

- Moderate rainfall forecast
- Potential localized flooding

Actions:

- Pre-position supplies at block-level warehouses
 - Alert local volunteers
 - Verify transport availability
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High Risk (5,000–8,000 kits)

Characteristics:

- Significant rainfall forecast
- High probability of relief operations

Actions:

- Mobilize district response teams
 - Arrange transportation contracts
 - Increase warehouse stock levels
 - Coordinate with district authorities
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Severe Risk (> 8,000 kits)

Characteristics:

- Extreme rainfall forecast
- High population exposure
- Elevated flood vulnerability

Actions:

- Immediate pre-positioning of supplies
 - Activate emergency response teams
 - Establish temporary storage hubs
 - Coordinate with government agencies and local administration
 - Prioritize vulnerable communities and low-accessibility areas
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Resource Pre-Positioning Strategy

Supplies should be moved before floodwaters disrupt transportation routes.

Recommended inventory:

- Dry ration kits
- Drinking water supplies
- ORS packets
- Tarpaulins
- Basic medical kits

Priority districts should receive supplies 48–72 hours before expected peak rainfall.

Recommended Decision Cycle

Frequency	Action
Daily	Review rainfall forecast and demand forecast
Every 3 Days	Re-evaluate stock levels
Weekly	Update district prioritization
During Flood Events	Monitor dashboard every 6–12 hours

Expected Benefits

- Faster response times
- Reduced stock shortages
- Better logistics planning
- Improved targeting of vulnerable populations
- More efficient use of NGO resources